

| Model         | F1 ↑ | DAF ↑ | Prec ↑ | Acc ↑ | TPR ↑ | TNR ↑ | MAE ↓ | RMSE ↓ | R <sup>2</sup> ↑ | Training Size  | Model Type |
|---------------|------|-------|--------|-------|-------|-------|-------|--------|------------------|----------------|------------|
| CHGNet        | 0.61 | 3.04  | 0.52   | 0.84  | 0.74  | 0.86  | 0.06  | 0.10   | 0.69             | 1.6M (145.9K)  | UIP-GNN    |
| M3GNet        | 0.58 | 2.65  | 0.45   | 0.80  | 0.79  | 0.80  | 0.07  | 0.12   | 0.58             | 188.3K (62.8K) | UIP-GNN    |
| M3GNet→MEGNet | 0.55 | 2.54  | 0.43   | 0.79  | 0.75  | 0.80  | 0.09  | 0.14   | 0.36             | 188.3K (62.8K) | UIP-GNN    |
| CHGNet→MEGNet | 0.55 | 2.56  | 0.43   | 0.79  | 0.74  | 0.80  | 0.09  | 0.15   | 0.34             | 1.6M (145.9K)  | UIP-GNN    |
| MEGNet        | 0.51 | 2.70  | 0.46   | 0.81  | 0.57  | 0.86  | 0.13  | 0.20   | -0.28            | 133.4K         | GNN        |